MIT ADT UNIVERSITY, PUNE
Department of Computer Science and Engineering
Generative AI · Mini Project · 2026–2027

DataPulse AI

Autonomous CSV Dataset Profiler, ML Task Recommender & Generative AI Studio

Problem Statement P10 · CSV Dataset Insight & Task Recommender

STUDENT DETAILS

STUDENT NAME
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10

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PROJECT BRIEF

DataPulse AI is an autonomous, production-grade tabular dataset profiler and machine learning task recommender built to streamline the entire exploratory data science lifecycle. Users upload any arbitrary CSV dataset to receive immediate structural insights, data health scoring (penalizing missingness, duplicates, and outliers), and heuristic ML problem classification (Supervised Regression, Binary or Multiclass Classification, or Unsupervised Clustering). The system integrates a sandboxed LLM code assistant that writes, strictly AST-validates, and runs pandas transformations with automatic narrative interpretations. To bridge multimodal and generative capabilities, columns are semantically clustered using Sentence-BERT embeddings, synthetic tabular samples are generated and distribution-validated using a PyTorch CTGAN generator, and automated Responsible AI audits proactively detect PII leaks and critical class imbalances.

KEY FEATURES

- Statistical column profiling & ML lifecycle recommendation (U1)
- Sandboxed LLM pandas code generation & safe AST execution (U2)
- Sentence-BERT column clustering & Vision LLM charts (U3)
- PyTorch CTGAN synthetic tabular generation & KS benchmark (U4)
- Responsible AI audit: PII detection & class imbalance (U5)
- Built with Python 3.12, PyTorch, and Streamlit Community Cloud

PROJECT LINKS

LIVE APPLICATION
https://datapulse-ai-p10.streamlit.app/

GITHUB REPOSITORY
https://github.com/atharva1145/Datapulse-ai-p10